

Part I

SUMMARY of Antiderivatives:

Summary

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A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$, for all x in I .

If F is an antiderivative of f on an interval I , then the function f has a whole **family of antiderivatives**.

If a function G is an antiderivative of f on I , then $G(x) = F(x) + C$, for all x in I , for some constant C .

The family of of *all* antiderivatives of f is denoted by $\int f(x) dx$ and called **indefinite integral of f** . Therefore, if F is an antiderivative of f , $\int f(x) dx = F(x) + C$.

Basic Indefinite Integrals

- $\int k dx = kx + C$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\ln(a)} + C$
- $\int \cos(x) dx = \sin(x) + C$
- $\int \sin(x) dx = -\cos(x) + C$
- $\int \sec^2(x) dx = \tan(x) + C$
- $\int \csc^2(x) dx = -\cot(x) + C$
- $\int \sec(x) \tan(x) dx = \sec(x) + C$
- $\int_C \csc(x) \cot(x) dx = -\csc(x) + C$
- $\int \frac{1}{x^2 + 1} dx = \arctan(x) + C$
- $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) + C$

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 Find the most general antiderivative of the function

$$g(t) = e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2}$$

Explanation. The family of all antiderivatives of g is given by

$$\int \left(e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2} \right) dt.$$

We can apply the Sum Rule and Constant Multiple Rule.

$$\int \left(e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2} \right) dt = \int e^{-2t} dt - 5 \int dt + 6 \int \sqrt{t} dt - 7 \int \frac{1}{t} dt + 5 \int \frac{1}{1+t^2} dt$$

And finally,

$$\int \left(e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2} \right) dt = -\frac{1}{2}e^{-2t} - 5t + 4t^{\frac{3}{2}} - 7 \ln |t| + 5 \arctan t + C.$$

Problem 2

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Problem 2 Determine the following indefinite integrals.

(a)

$$\int (\sec^2(x) + 5) dx$$

Explanation.

$$\int (\sec^2(x) + 5) dx = \tan(x) + 5x + C$$

(b)

$$\int (\sec^2(x) - \sec(x) \tan(x)) dx$$

Explanation. We apply the Sum Rule.

$$\int (\sec^2(x) - \sec(x) \tan(x)) dx = \int \sec^2(x) dx - \int \sec(x) \tan(x) dx = \tan(x) - \sec(x) + C.$$

(c)

$$\int \frac{x + 3x^5}{x^3} dx$$

Explanation. First, we express the function in a more convenient way.

$$\int \frac{x + 3x^5}{x^3} dx = \int \left(\frac{x}{x^3} + \frac{3x^5}{x^3} \right) dx = \int (x^{-2} + 3x^2) dx.$$

Now we apply the Sum Rule and the Constant Multiple Rule.

$$\int \frac{x + 3x^5}{x^3} dx = \int (x^{-2} + 3x^2) dx = \int x^{-2} dx + 3 \int x^2 dx.$$

Now we determine each integral.

$$\int \frac{x + 3x^5}{x^3} dx = \int x^{-2} dx + 3 \int x^2 dx = -\frac{1}{x} + x^3 + C.$$

(d)

$$\int \frac{1+2x}{1+x^2} dx$$

(HINT: Think about the derivative of $\ln(f(x))$).

Explanation. First, we express the function in a more convenient way.

$$\int \frac{1+2x}{1+x^2} dx = \int \left(\frac{1}{1+x^2} + \frac{2x}{1+x^2} \right) dx.$$

Now we apply the Sum Rule.

$$\int \frac{1+2x}{1+x^2} dx = \int \frac{1}{1+x^2} dx + \int \frac{2x}{1+x^2} dx.$$

Now we determine each integral.

$$\int \frac{1+2x}{1+x^2} dx = \int \frac{1}{1+x^2} dx + \int \frac{2x}{1+x^2} dx = \arctan(x) + \ln(1+x^2) + C.$$

(e)

$$\int \frac{2+x^2}{1+x^2} dx$$

Explanation. First, we express the function in a more convenient way.

$$\int \frac{2+x^2}{1+x^2} dx = \int \frac{1+1+x^2}{1+x^2} dx = \int \left(\frac{1}{1+x^2} + \frac{1+x^2}{1+x^2} \right) dx = \int \left(\frac{1}{1+x^2} + 1 \right) dx.$$

Now we apply the Sum Rule and the Constant Multiple Rule.

$$\int \frac{2+x^2}{1+x^2} dx = \int \frac{1}{1+x^2} dx + \int dx = \arctan(x) + x + C.$$

Problem 3

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Problem 3

Problem 3 Assume that $f'(t) = 4t^3 + 2t$ and $f(3) = 5$. Find $f(t)$.

Explanation. This is an Initial Value Problem. First we find the general solution of the differential equation.

$$f(t) = t^4 + t^2 + C.$$

Then, we find the unique solution of the IVP.

$$5 = f(3) = 3^4 + 3^2 + C = 81 + 9 + C = 90 + C$$

$$\implies C = -85$$

and so

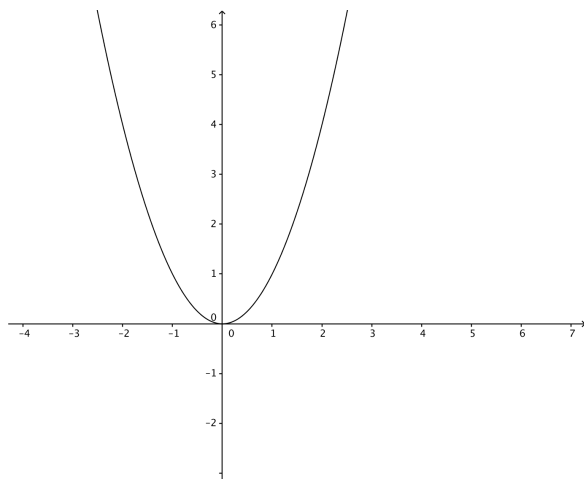
$$f(t) = t^4 + t^2 - 85.$$

Problem 4

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Problem 4

Problem 4 The graph of a function f is the parabola given in the figure below.

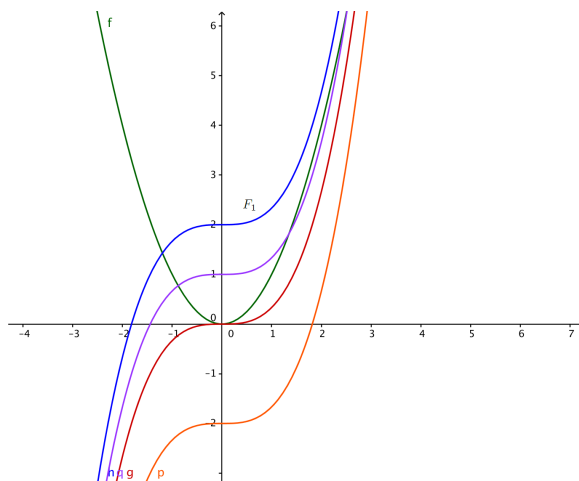


- (a) Find a formula for $f(x)$.

Explanation. $f(x) = x^2$

- (b) Suppose F_1 is an anti-derivative of f satisfying $F_1(0) = 2$. Sketch and label the graph of F_1 .
Sketch the graphs of three more antiderivatives of f .

Explanation.



- (c) Using the expression you found in part (a) and your sketches in part (b), find the algebraic representation of F_1 .

Problem 4

Explanation. The general antiderivative is $F(x) = \frac{1}{3}x^3 + C$. Therefore, $F_1(x) = \frac{1}{3}x^3 + C$, for some constant C .

Since $F_1(0) = 2$, it follows that $2 = \frac{1}{3}(0)^3 + C = C$. Therefore, $F_1(x) = \frac{1}{3}x^3 + 2$.

- (d) Suppose we're given $h(x) = (1/3)x^3 + 17$, what is $h'(x)$?

Explanation. $h'(x) = x^2$.

- (e) What is the relationship between f , F_1 , h , and h' ?

Explanation. We have $F_1' = f = h'$ and $h = F_1 + C$, where C is some constant. Determine this constant!

Problem 5

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Problem 5 Given the acceleration function, the initial velocity and initial position of an object moving along a line, find the **position function**.

$$a(t) = 4 \cos t, v(0) = 2, s(0) = 6.$$

Explanation. $v(t) = 4 \sin t + C$.

Letting $t = 0$, we get

$$v(0) = 4 \sin 0 + C \text{ and}$$

$$2 = C.$$

Therefore, $v(t) = 4 \sin t + 2$.

So, $s(t) = -4 \cos t + 2t + C$.

Letting $t = 0$, we get

$$s(0) = -4 \cos 0 + 2(0) + C.$$

So,

$$6 = -4 + C,$$

and $C = 10$.

Therefore,

$$s(t) = -4 \cos t + 2t + 10.$$

Problem 6

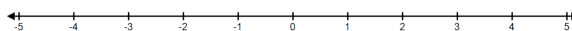
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Problem 6 Consider an object moving along a line with velocity $v(t) = \pi \sin(\pi t)$ on $[0, 2]$ and initial position $s(0) = 0$. Time is measured in seconds and velocity in m/s.

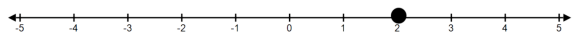
- (a) Determine the position function, $s(t)$, on $[0, 2]$.

Explanation. General antiderivative of $v(t)$: $s(t) = -\cos(\pi t) + C$
 Apply initial position: $s(t) = -\cos(\pi t) + C \implies 0 = s(0) = -\cos(0) + C \implies C = 1$
 Position function: $s(t) = -\cos(\pi t) + 1$

- (b) Mark the position of the object at the time $t = 1$ on the line below.



Explanation. $s(1) = -\cos(\pi) + 1 = -(-1) + 1 = 2$



- (c) Determine the average velocity, v_{av} , of the object during the interval $[0, 2]$.

Explanation.

$$\begin{aligned} v_{av} &= \frac{s(2) - s(0)}{2 - 0} \\ &= \frac{-\cos(2\pi) + 1 - (-\cos(0) + 1)}{2} \\ &= \frac{-\cos(2\pi) + \cos(0)}{2} \\ &= \frac{-1 + 1}{2} = 0 \end{aligned}$$

- (d) Determine when the motion is in the positive direction.

Explanation. Motion is in the positive direction when $v(t) > 0$.

$$\begin{aligned} v(t) > 0 &\iff \pi \sin(\pi t) > 0 \\ &\iff \sin(\pi t) > 0 \\ &\implies 0 < \pi t < \pi \\ &\iff 0 < t < 1 \end{aligned}$$

- (e) At what time (or times) is the object farthest from the origin?

Problem 6

Explanation. Object is farthest from the origin when $|s(t) - 0| = |1 - \cos(\pi t)|$ is maximized. Since $s(t) \geq t$ for t in $[0, 2]$ we need to maximize $s(t)$. Finding critical points on the open interval $(0, 2)$.

$$\begin{aligned}v(t) = 0 &\iff \pi \sin(\pi t) = 0 \\&\implies \pi t = \pi \\&\iff t = 1\end{aligned}$$

Finding the absolute maximum, we have to check the endpoints and the critical points:

$$\begin{aligned}s(0) &= 0 \\s(1) &= 2 \\s(2) &= 0\end{aligned}$$

Hence the object is farthest from the origin at $t = 1$.

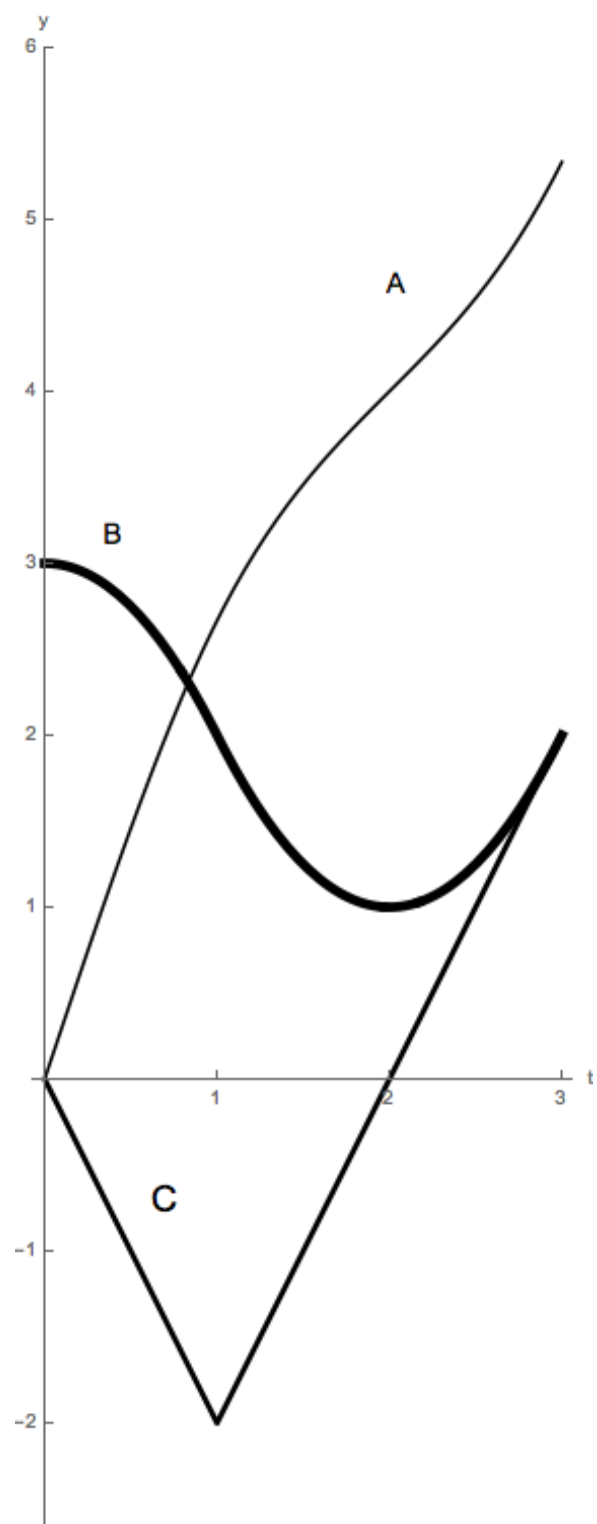
Problem 7

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Problem 7

Problem 7 Consider an object moving along a straight line. The graphs of acceleration function (in m/s^2), the velocity function and the position function of the object are given in the figure below.

Problem 7



Problem 7

Find the initial velocity, $v(0)$, and initial position, $s(0)$, of the object. (HINT: Start by determining which graph corresponds to each of $s(t)$, $v(t)$, and $a(t)$.)

Explanation. It is clear that the function B is the velocity. Therefore, $v(0) = 3$. It is clear that the function A is the position function. Therefore, $s(0) = 0$.